ZeroFault API — how it works

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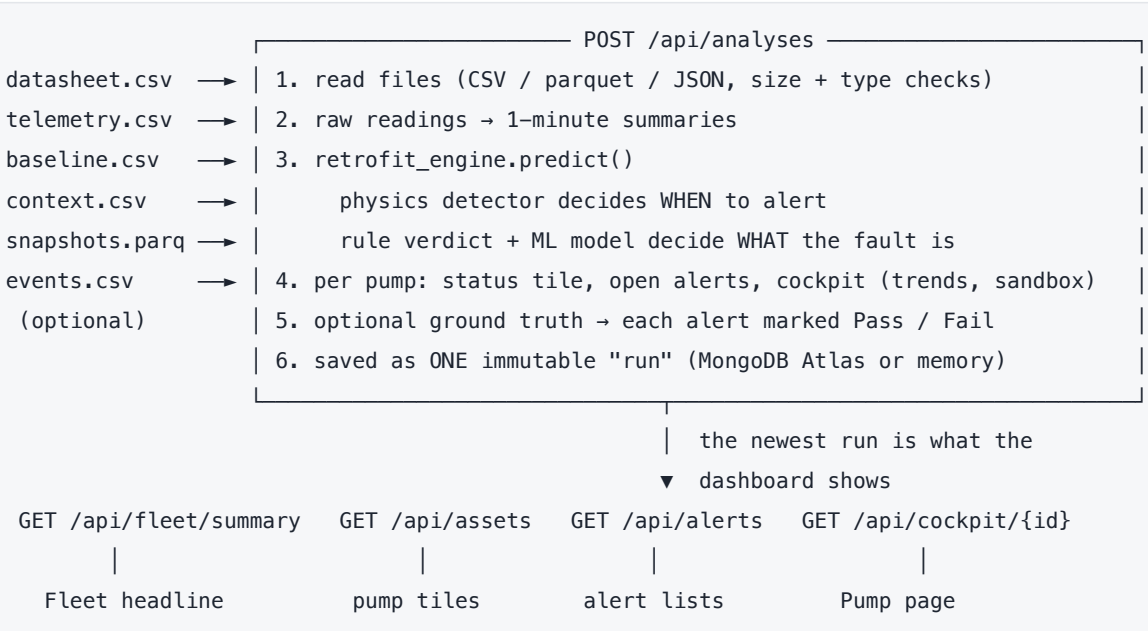
The REST API behind the operator dashboard. It runs our trained hybrid model (Joice's physics rules + the ML diagnostician in `retrofit_engine/`) on uploaded pump data, stores each result, and serves it to the Fleet and Pump screens.

- Base URL (local): `http://localhost:8000`
- Interactive docs: `http://localhost:8000/docs` (Swagger), `http://localhost:8000/redoc`
- All responses are JSON unless noted. Times are ISO-8601 UTC (`2025-08-12T19:27:00Z`).
- The API is **read-only towards the pumps**: it never writes to a PLC or changes a setting.

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1. The big picture



Key ideas:

Idea	What it means
Run	One model execution over one set of uploaded files. Has a `run_id`. Never changed after it is saved.
Latest run	All dashboard endpoints read the most recent run. A new upload instantly becomes what everyone sees.
Seed run	On startup, if the store is empty, `server/seed/demo_run.json` is loaded so the dashboard is never blank.

Idea	What it means
Open alert	Any alert of the pump in the analysed period (same alerts as the "Detected issues" card). Optionally limited by `ACTIVE_ALERT_HOURS`. Drives the pump's status.
Severity	`act_now` · `attention` · `sensor` · `ok` — from the urgency of the pump's most severe open alert.

2. Code map

```
server/src/
├── main.py           app, CORS, lifespan (open store, load seed), /health, mounts routers
├── settings.py       environment variables → Settings
├── engine.py         imports retrofit_engine (repo root); API still starts if it is missing
├── analysis.py       run_analysis(): calls the model and builds tiles / cockpits / CSV
├── sandbox.py        what-if speed derate (pump affinity laws)
├── fault_guide.py    plain names, urgency, steps per fault (data/fault_actions.json)
├── db.py             RunStore: MongoRunStore (Atlas) or MemoryRunStore
├── deps.py           FastAPI dependencies (store, settings) — overridden in tests
├── models.py         Pydantic request/response models
├── sample_data.py    static retrofit scorecard
└── routers/
    ├── dashboard.py  /api/alerts, /api/assets, /api/fleet/summary, /api/cockpit, /api/fault-guide,
/api/scorecard, /api/model
    └── analyses.py   /api/analyses (upload + run list + CSV export)
```

3. Endpoints at a glance

Method	Path	Used by	Purpose
GET	`/health`	App start-up	Is the API up, is the DB connected, which model is loaded
GET	`/api/fleet/summary`	Fleet screen	Headline counts
GET	`/api/assets`	Fleet screen	One status tile per pump
GET	`/api/alerts`	Fleet + Pump screens	Alerts of the latest run (filterable)
GET	`/api/alerts/{asset_id}`	—	Alerts of one pump (404 if none)
GET	`/api/cockpit/{asset_id}`	Pump screen	Diagnosis, evidence, steps, trends, sandbox
GET	`/api/fault-guide`	Pump screen ("What happens next?")	What each fault means, how urgent, what to do
GET	`/api/scorecard`	Scorecard screen	Retrofit business case (static)
GET	`/api/model`	Upload dialog	Model version, required columns, upload limit
POST	`/api/analyses`	Upload dialog	Upload files → run model → new latest run
GET	`/api/analyses`	—	Recent runs, newest first
GET	`/api/analyses/latest`	App (run strip)	The run the dashboard is showing
GET	`/api/analyses/{run_id}`	—	One run's summary

Method	Path	Used by	Purpose
GET	`/api/analyses/{run_id}/alerts.csv`	—	Alerts in the organisers' template CSV
GET	`/retrofit/info`	—	Kaleb's stateless engine info
POST	`/retrofit/analyze`, `/retrofit/analyze/csv`	—	Kaleb's stateless engine (result not stored)

4. Endpoint details

`GET /health`

```
{ "status": "ok", "service": "since-ai-server", "database": "connected", "model": "hybrid-v3-001" }
```

`database` is `connected` (Atlas) or `disabled` (memory). `model` is `unavailable` when `retrofit_engine` could not load — dashboard reads still work, uploads return 503.

`GET /api/fleet/summary`

```
{
  "vessel": "4 sites · Fleet overview",
  "watch": "Model hybrid-v3-001 · read-only",
  "critical": 2,
  "actionable": 6,
  "sensor_faults": 1,
  "availability_pct": 22.2,
  "healthy": 2,
  "total": 9,
  "nuisance_suppressed_pct": null,
  "run_id": "e001660ed19b433db16a30d8e1e01ec4",
  "model_version": "hybrid-v3-001",
  "data_end_utc": "2025-08-18T00:00:00Z"
}
```

Field	Rule
`critical`	pumps with severity `act_now`
`actionable`	`act_now` + `attention`
`sensor_faults`	pumps whose worst open alert is a sensor problem
`availability_pct`	`healthy / total × 100`
`nuisance_suppressed_pct`	not computed by the model yet → `null`

`GET /api/assets`

One tile per pump in the latest run.

```
[
  {
    "asset_id": "PUMP-01",
    "site": "Vessel A (RoPax, fictional)",
    "system": "Seawater cooling, pump No. 1",
    "tier": "T1",
    "status": "alert",
    "mode": "RUNNING",
    "load_pct": 19.9,
    "open_alerts": 2,
    "fault_summary": "Inlet filter is blocked",
    "rul_hours": null,
    "severity": "attention",
    "last_data_utc": "2025-08-18T00:00:00Z"
  }
]
```


How each field is filled:

Field	Source
`site`, `system`	datasheet columns `site`, `system` (fallback `profile`, then `<control> pump`)
`tier`	`T0` / `T1` / `T2` from the datasheet or the sensors present in the telemetry
`mode`	most common `operating_mode` in the pump's last 15 minutes
`load_pct`	power / rated kW → else current / rated A → else speed / rated rpm (0 when not running)
`open_alerts`	the pump's alerts in the analysed period (or within `ACTIVE_ALERT_HOURS` of `last_data_utc` when set)
`severity`	urgency of the most severe open alert (see §6); `ok` when none
`status`	`alert` (act_now / attention) · `sensor_issue` (sensor) · `healthy` (ok) — kept for older UI code
`fault_summary`	plain name of the most severe open alert
`rul_hours`	`null` — the model does not estimate remaining life

`GET /api/alerts`

Query parameters (both optional):

Name	Values
`asset_id`	e.g. `PUMP-07`
`event_class`	`MACHINE` or `SENSOR` (anything else → 422)
`severity`	repeatable: `SEVERE`, `MODERATE`, `MINIMAL`, `LOW_CONFIDENCE`, e.g. `?severity=SEVERE&severity=MODERATE`

```
[
  {
    "asset_id": "PUMP-01",
    "alert_time_utc": "2025-08-12T19:27:00Z",
    "event_class": "MACHINE",
    "fault_type": "SUCTION_STRAINER_CLOGGING",
    "confidence": 0.827,
    "evidence": "suction_pressure_kpa z=-6.4; head +0% power -4% vs reference | ML agrees (0.83)",
    "recommended_action": "Inspect / clean suction strainer.",
    "true_fault_type": "SUCTION_STRAINER_CLOGGING",
    "correct": true,
    "severity": "SEVERE"
  }
]
```


`severity` is a per-alert triage band derived by the server (`src/severity.py`, in the shared contract): `SEVERE` = MACHINE alert with confidence ≥ 0.75 · `MODERATE` = other alerts ≥ 0.50 · `MINIMAL` = below 0.50 · `LOW_CONFIDENCE` = fault type `UNKNOWN`. It never changes the detector's fields or the benchmark CSV. Not to be confused with a pump's `severity` in `/api/assets` (act_now / attention / sensor / ok).

- Shape = `AlertPrediction` from `shared/contracts.schema.json` (+ the existing optional `true_fault_type`, `correct`, filled only when ground truth was uploaded).
- Sorted by pump, then time (oldest first).
- `fault_type` is one of the 13 benchmark faults or `UNKNOWN`.
- `confidence` is the calibrated model confidence (0–1). When the ML model agrees with the physics rule it is the higher of the two; when the ML model overrides the rule it is the ML confidence.

`GET /api/alerts/{asset_id}` returns the same list for one pump, or **404** if it has no alerts.

`GET /api/cockpit/{asset_id}`

Everything the Pump page needs. **404** for an unknown pump.

```
{
  "asset_id": "PUMP-01",
  "title": "Seawater cooling, pump No. 1",
  "diagnosis": "Inlet filter is blocked",
  "fault_type": "SUCTION_STRAINER_CLOGGING",
  "event_class": "MACHINE",
  "confidence": 0.75,
  "severity": "SEVERE",
  "safe_window": "Hours to days. Act now if cavitation alarms appear on the same pump.",
  "evidence": [
    { "channel": "Suction pressure", "sigma": -9.4, "note": "Below this pump's own normal" }
  ],
  "sop": [
    "Switch to the standby pump if one is available.",
    "Inspect and clean the suction strainer.",
    "After cleaning, check that suction pressure is back to normal."
  ],
  "trends": [
    { "name": "Suction pressure", "unit": "kPa", "threshold": null,
      "points": [ { "t": "2025-08-12T06:00:00Z", "value": 48.21 }, "… up to 168 hourly points" ] }
  ],
  "sandbox_default_speed_pct": 100,
  "npsh_at_100": 11.95,
  "npsh_at_88": 12.27,
  "risk_at_100": 0.75,
  "risk_at_88": 0.582,
  "energy_delta_eur_day": -12.88,
  "cooling_margin_pct": 110,
  "cooling_safe_flow_m3h": 252.4,
  "rated_rpm": 1780,
  "sandbox_note": "Affinity-law estimate from the last running hour. … Assumed: no flow meter: flow
estimated from speed."
}
```


Part	How it is built
`diagnosis`, `fault_type`	the most severe open alert (a known fault beats UNKNOWN, then most recent); `"No active fault"` when none
`safe_window`, `sop`	urgency text and steps from `fault_guide` for that fault
`evidence`	channel z-scores parsed from the alert's evidence text (`sigma` = deviations from the pump's own normal)
`trends`	up to 3 series: the channels named in the evidence first, then defaults (vibration/bearing for mechanical faults, pressures/flow/power for hydraulic ones). Hourly means, ≤ 168 points. `threshold` = datasheet `vibration_alarm_mm_s` for vibration RMS.
sandbox fields	see §7

`GET /api/fault-guide`

Plain-language guidance for all 14 fault types, from `src/data/fault_actions.json` (copy of `joice_physics_rules/dashboard_data/fault_actions.json`). The pump page's **"What happens next?"** tab shows one card per issue that failed on the pump, most urgent first.

```
[
  {
    "fault_type": "BEARING_WEAR",
    "event_class": "MACHINE",
    "plain_name": "Bearing is wearing out",
    "what_it_means": "The ball bearing that carries the shaft is damaged. …",
    "what_to_do": ["Plan a bearing replacement at the next stop.", "Check lubrication now (grease
level, correct grease).", "…"],
    "urgency_level": "next_planned_stop",
    "urgency_text": "Days to weeks. Becomes urgent if bearing temperature keeps rising.",
    "not_to_confuse_with": "Cavitation also shakes the pump, but then the bearing index stays near
normal …"
  }
]
```

`urgency_level` colours (as defined in the file): `act_now` red · `within_a_day` orange · `next_planned_stop` yellow · `check_sensor` blue.

`GET /api/scorecard`

Static retrofit business case (BOM, economics, T0 vs T1 capabilities). Not computed from runs.

`GET /api/model`

```
{
  "available": true,
  "model_version": "hybrid-v3-001",
  "physics_version": "joice-v3",
  "classes": ["AIR_ENTRAINMENT", "BEARING_WEAR", "…", "NO_FAULT", "…"],
  "calibrated": true,
  "reason": null,
  "required_datasheet_columns": ["asset_id", "control", "motor_rated_rpm", "pump_rated_flow_m3_h",
                                 "pump_rated_head_m", "pump_shutoff_head_m", "npsh3_at_bep_m",
"fluid"],
  "min_history_hours": 24,
  "max_upload_mb": 200
}
```

When the engine cannot load: `{"available": false, "reason": "<why>", "max_upload_mb": 200, …}`.

`POST /api/analyses` — upload and run the model

`multipart/form-data`:

Field	Required	Format	Content
`datasheet`	**yes**	csv / parquet / json	one row per pump; required columns from `/api/model`. Optional: `site, system, profile, motor_rated_kw, motor_rated_current_a, bearing_bpfo_order, impeller_vanes, vibration_alarm_mm_s, min_speed_ratio, instrumentation_tier, channels_available`
`telemetry`	**yes**	csv / parquet / json	the period to check. Either raw readings `asset_id, timestamp_utc, <sensor columns>` (any rate) or 1-minute summaries `asset_id, window_start_utc, <sensor>__mean/std/min/max/count, operating_mode, running_fraction, rows`
`baseline`	no	same as telemetry	≥ 24 h of healthy running just before the telemetry. Without it the first ~24 h of telemetry are "normal" and get no alerts.
`context`	no	csv	`asset_id, timestamp_utc, sea_state_hs_m, suction_static_head_m`
`snapshots`	no	parquet	vibration waveforms `asset_id, timestamp_utc, sample_rate_hz, rpm_at_capture, samples` (improves bearing / misalignment / imbalance)
`events`	no	csv / json	ground truth `asset_id, fault_type, onset_utc[, end_utc]` → alerts get `true_fault_type` + `correct`
`use_ml`	no	form field	`true` (default) = rules + ML; `false` = physics rules only

Sensor columns: `motor_speed_rpm, motor_current_a, motor_power_kw, suction_pressure_kpa, discharge_pressure_kpa, flow_m3_h, fluid_temp_c, bearing_temp_c, motor_winding_temp_c, vibration_rms_mm_s, vibration_accel_rms_g, vibration_crest_factor, ambient_temp_c, valve_position_pct` — send the ones the pump has.

**Response `201 Created` ** — the run summary (it is now the latest run):


```
{
  "run_id": "e6a188e005694c14bdc1a7540e3d6bb4",
  "created_utc": "2026-11-07T19:30:43Z",
  "source": "upload",
  "label": "datasheet.csv + telemetry.csv",
  "model_version": "hybrid-v3-001",
  "physics_version": "joice-v3",
  "use_ml": true,
  "pumps": 2,
  "alerts": 13,
  "ml_overrides": 1,
  "by_fault_type": { "MISALIGNMENT": 3, "SUCTION_STRAINER_CLOGGING": 2, "BEARING_WEAR": 2, "…": 1 },
  "data_start_utc": "2025-09-09T06:00:00Z",
  "data_end_utc": "2026-02-16T00:00:00Z",
  "processing_seconds": 0.75,
  "scored": false,
  "notes": ["datasheet: instrumentation_tier derived from the sensors present in the telemetry"]
}
```


`notes` are warnings worth showing (short baseline, missing snapshots, derived columns, pumps skipped…).

Errors — body is always `{"detail": "<message for the user>"}`:

Status	When	Example `detail`
415	file is not csv / parquet / json	`telemetry file 'data.xlsx': use CSV, parquet or JSON`
413	a file is over `MAX_UPLOAD_MB`	`telemetry file 'big.csv' is larger than 200 MB`
422	empty / unreadable file, missing or bad columns, unknown control type, no pump in both files	`datasheet is missing columns ['control', …]; required: […]`
422	a required form field is missing	FastAPI validation list
503	the model engine is not installed / failed to load	`model engine unavailable: …`

Typical time: under 1 s for 2 pumps × 5 days of 1-minute data, ~3 s for 9 pumps × 7 days. Analyses run one at a time (a lock) so parallel uploads cannot starve the API; raw 1 Hz files take longer because they are aggregated first.

Run history

- `GET /api/analyses?limit=20` → list of run summaries, newest first (`limit` 1–100).
- `GET /api/analyses/latest` → the run the dashboard shows (404 before any run exists).
- `GET /api/analyses/{run_id}` → one summary (404 if unknown).
- `GET /api/analyses/{run_id}/alerts.csv` → `text/csv` download with the organisers' template columns `asset_id,alert_time_utc,event_class,fault_type,confidence,evidence,recommended_action`.

5. Examples

curl — upload and read back

```
curl -X POST http://localhost:8000/api/analyses \
  -F datasheet=@datasheet.csv -F telemetry=@telemetry.csv -F baseline=@baseline.csv \
  -F context=@context.csv -F snapshots=@snapshots.parquet -F use_ml=true

curl http://localhost:8000/api/fleet/summary
curl "http://localhost:8000/api/alerts?asset_id=PUMP-09&event_class=MACHINE"
curl http://localhost:8000/api/cockpit/PUMP-09
curl -OJ http://localhost:8000/api/analyses/<run_id>/alerts.csv
```

Demo upload files: `python -m retrofit_engine.tools.make_demo_files --data <dataset>/public --pumps PUMP-08 PUMP-09 --out demo_files`.

TypeScript (client/src/api/client.ts)

```
const form = new FormData();
form.append("datasheet", datasheetFile);
form.append("telemetry", telemetryFile);
form.append("use_ml", "true");
const run = await createAnalysis(form);         // POST /api/analyses → AnalysisRun
const cockpit = await fetchCockpit("PUMP-09");  // reads the new latest run
// errors throw ApiError { status, message = server's "detail" }
```

6. Severity rules

Each fault type has an urgency level in `src/data/fault_actions.json` (Joice's guide). The pump takes the level of its most severe open alert:

Urgency level	Faults	Dashboard severity
`act_now`	CAVITATION, AIR_ENTRAINMENT	**act_now** (red)
`within_a_day`	SUCTION_STRAINER_CLOGGING, DISCHARGE_RESTRICTION, UNKNOWN	**attention** (orange)
`next_planned_stop`	BEARING_WEAR, MISALIGNMENT, IMBALANCE, IMPELLER_WEAR	**attention**
`check_sensor`	SENSOR_DRIFT, SENSOR_STUCK, SENSOR_CALIBRATION_SHIFT, SENSOR_NOISE_SPIKES, COMMUNICATION_LOSS	**sensor** (blue)
no open alert	—	**ok** (green)

7. Mitigation sandbox (cockpit "What happens next?")

Computed in `src/sandbox.py` from the median of the pump's last running hour and its datasheet, for today's speed (100 %) and 88 % of it. The UI slider interpolates between the two.

Field	Physics
`npsh_at_100`, `npsh_at_88`	NPSH available − NPSH required, in metres. NPSHr ∝ speed²; NPSHa from suction pressure, water vapour pressure (Antoine) and velocity head. Higher = further from cavitation.
`energy_delta_eur_day`	power ∝ speed³ → `(P×0.88³ − P) × 24 h × ENERGY_PRICE_EUR_KWH` (negative = saving)
`cooling_margin_pct`	flow at 88 % speed vs the share of today's flow the cooled system needs (`COOLING_MIN_FLOW_FRACTION`)
`cooling_safe_flow_m3h`	that minimum flow
`risk_at_100`	model confidence that the machine fault is present (0.05 when healthy / sensor-only)
`risk_at_88`	suction-driven faults scaled by NPSH ratio; bearing / misalignment / imbalance × 0.88²; others × 0.88

Risk is a guide, not a failure probability. `sandbox_note` states this and lists per-pump assumptions (no flow meter, no suction gauge, fixed-speed pump without a drive, 88 % below the drive minimum…).

8. Storage

Mode	When	Behaviour
MongoDB Atlas	`MONGODB_URI` set and reachable	collection `analysis_runs`, one document per run (`_id` = `run_id`), indexed by `summary.created_utc`; recent runs cached in memory
Memory	`MONGODB_URI` empty or Atlas unreachable	runs live until the process restarts

On startup, if there is no run yet, the seed run (`SEED_RUN_PATH`) is saved. Rebuild it after a model update: `python -m scripts.build_seed_run --data <dataset>/public` (see `README.md`).

9. Settings

Variable	Default	Meaning
`MONGODB_URI`, `MONGODB_DB`	empty, `since_ai`	Atlas connection
`CORS_ORIGINS`	localhost + VM IP	comma-separated allowed origins
`MAX_UPLOAD_MB`	200	per uploaded file (the host nginx `client_max_body_size` must allow it too)
`ACTIVE_ALERT_HOURS`	empty	empty = whole analysed period; a number = only alerts that many hours before the last reading
`ENERGY_PRICE_EUR_KWH`	0.15	sandbox energy cost
`COOLING_MIN_FLOW_FRACTION`	0.8	sandbox cooling requirement
`SEED_RUN_PATH`	`seed/demo_run.json`	demo run for an empty store

10. Troubleshooting

Symptom	Cause / fix
Dashboard shows "example data" banner	`/health` not reachable — API down or wrong `VITE_API_BASE_URL` / proxy
`/health` says `"model": "unavailable"`	`retrofit_engine/` missing or a dependency (lightgbm needs `libgomp1`) failed; see startup log `Model engine: unavailable — …`
Upload returns 413 from nginx (HTML page, not JSON)	raise `client_max_body_size` on the host nginx
A pump has no alerts at all	less than ~24 h of data and no baseline — the model needs a "normal" period first
Bearing / misalignment rarely detected	upload `snapshots` (vibration waveforms)
`"database": "disabled"` in production	`MONGODB_URI` secret missing or Atlas unreachable — runs are lost on restart